

Euclid's Elements — Book XI

Proposition 27

Formal Reconstruction — Technical Documentation

Jurek Róžański
Military University of Technology, Warsaw, Poland
`jerzy.rozanski@wat.edu.pl`

CGJTEAMLAB

To describe a parallelepipedal solid similar and similarly situated to a given parallelepipedal solid on a given straight line.

1 Purpose and Scope

Proposition XI.27 is the constructive similarity theorem for parallelepipedal solids. Given a parallelepipedal solid and a prescribed nondegenerate straight line, Euclid constructs on that line a second parallelepipedal solid which is similar and similarly situated to the first one [1].

The proposition is structurally important because it combines four layers which had previously been developed largely independently:

1. spatial congruence of solid angles, supplied by XI.26;
2. planar fourth-proportional constructions, corresponding to VI.12;
3. composition of proportions by V.22;
4. similarity of parallelepipedal solids in the sense of XI.Def.9.

In the production formalization these layers are joined without introducing coordinates, vectors, scalar products, numerical angle measures, or a numerical scale factor. The construction remains synthetic throughout. The target solid is built from a copied trihedral corner, two synthetic fourth-proportional constructions, a third proportion obtained by V.22, and a parallel-plane shell completion.

The direct production theorem is

```
Geometry.euclid_proposition_11_27
```

in

```
CGJteamLab/Proposition11_27.lean
```

The supporting production modules which are particularly visible in XI.27 are

```
CGJteamLab/Proposition11_26.lean  
CGJteamLab/Proposition5_22.lean  
CGJteamLab/HilbertInterfaceVI.lean  
CGJteamLab/HilbertInterfaceXI.lean
```

The Book IV circle dependency is now transitive rather than a direct XI.27 import. The raw spatial V.22 theorem in `Proposition5_22.lean` uses the public crossing-rays endpoint

```
hilbert_crossing_rays_transfer
```

exported by

```
CGJteamLab/Proposition4_20.lean
```

whose dependency closure contains the required earlier Forder IV.12–IV.19 propositions and the reusable infrastructure in `HilbertInterfaceIV.lean`.

Unlike XI.26, the current production repository contains no separate Hilbert–Wyler proposition file for XI.27. The present documentation therefore describes the single audited production route.

A central architectural point is that the final XI.27 theorem does not assume a global ambient

```
HilbertOrder Geo
```

Planar order arguments are confined to explicit plane slices when they are needed.

2 Euclid’s Statement

Euclid’s Proposition XI.27 states:

To describe a parallelepipedal solid similar and similarly situated to a given parallelepipedal solid on a given straight line.

Classically, let AB be the prescribed straight line and let the given parallelepipedal solid have a trihedral corner at C . Euclid first uses XI.26 to copy the solid angle at C to the point A on the prescribed line. The two remaining target edge lengths are then chosen so that the three edges incident with the new vertex have the same mutual ratios as the three corresponding source edges.

The formal theorem uses an explicit source parallelepiped configuration

```
HilbertParallelepipedConfiguration
```

```

(Geo := Geo)
pi0 pi1 rho0 rho1 sigma0 sigma1
A B C D E F G Hpt

```

and a prescribed nondegenerate target edge A_0B_0 .

Its conclusion constructs points

$$K, J, TD, TE, TG, TH$$

and six target planes

$$t\pi_0, t\pi_1, \quad t\rho_0, t\rho_1, \quad t\sigma_0, t\sigma_1$$

such that the target eight-vertex configuration

$$(B_0, A_0, K, TD, TE, J, TG, TH)$$

is a parallelepiped and is similar to the source in the project representation of XI.Def.9.

The public theorem has the form

```

theorem euclid_proposition_11_27
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (pi0 pi1 rho0 rho1 sigma0 sigma1 : S.Plane)
  (A B C D E F G Hpt : Geo.Point)
  (hCfg :
    HilbertParallelepipedConfiguration
      (Geo := Geo)
      pi0 pi1 rho0 rho1 sigma0 sigma1
      A B C D E F G Hpt)
  (A0 B0 : Geo.Point)
  (hA0B0 : Ne A0 B0) :
  exists K J : Geo.Point,
  exists tpi0 tpi1 trho0 trho1 tsigma0 tsigma1 : S.Plane,
  exists TD TE TG TH : Geo.Point,
  HilbertParallelepipedConfiguration
    (Geo := Geo)
    tpi0 tpi1 trho0 trho1 tsigma0 tsigma1
    B0 A0 K TD TE J TG TH /\
  HilbertXI9SimilarParallelepiped
    (HilbertSpaceFaceCornerSimilarFace
      (Geo := Geo))
    (hilbertXI27FaceCorners
      (Geo := Geo)
      A B C D E F G Hpt)
    (hilbertXI27FaceCorners
      (Geo := Geo)
      B0 A0 K TD TE J TG TH)

```

Thus the theorem constructs the target solid rather than merely asserting the existence of an abstract similarity relation.

3 Classical Configuration

Euclid begins with a given parallelepipedal solid and chooses one of its vertices as the source trihedral corner. In the project notation used by the formal proof this source corner is the corner at B , with incident edges

$$BA, \quad BC, \quad BF.$$

The prescribed target edge is A_0B_0 . Proposition XI.26 is used first to reproduce at A_0 a trihedral angle congruent to the source trihedral angle at B . Before scaling, write the two newly copied rays as

$$A_0L, \quad A_0H_1.$$

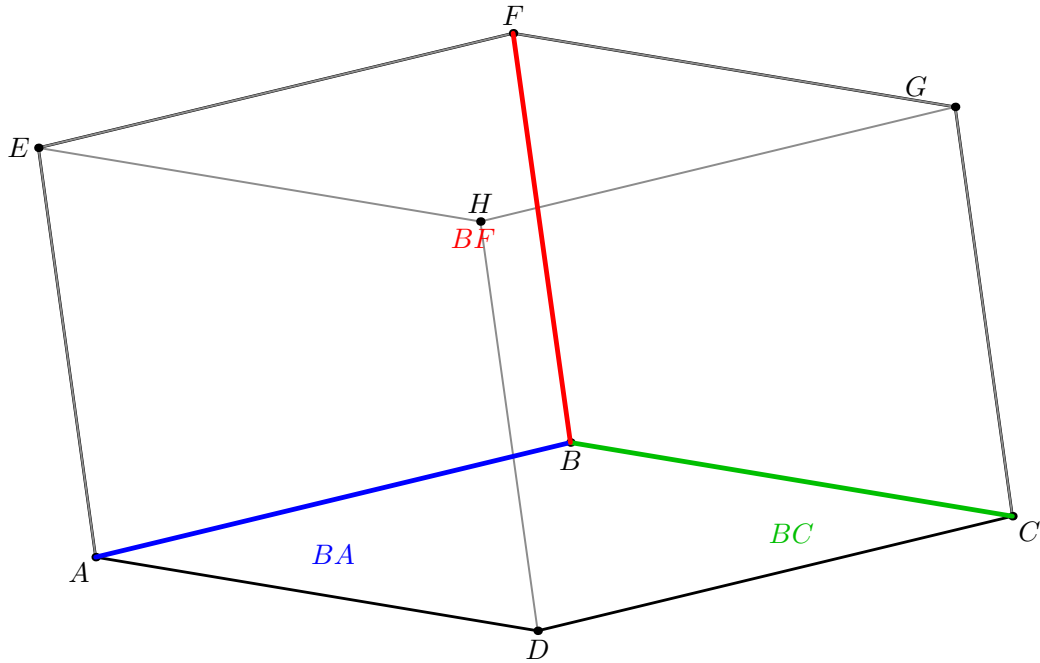


Figure 1: The source parallelepiped and the trihedral corner at B used by the formal construction. The three incident source edges are BA , BC , and BF ; the remaining vertices determine the three opposite faces.

The copied solid angle gives the three angular correspondences

$$\begin{aligned} \angle B_0A_0L &\cong \angle ABC, \\ \angle LA_0H_1 &\cong \angle CBF, \\ \angle H_1A_0B_0 &\cong \angle FBA. \end{aligned}$$

The lengths on the copied rays are not yet correct. They must be chosen so that the target incident edges have the same ratios as the corresponding source edges. Euclid therefore uses VI.12 twice.

First choose K on the ray A_0L so that

$$BA : BC = A_0B_0 : A_0K.$$

Then choose J on the ray A_0H_1 so that

$$BC : BF = A_0K : A_0J.$$

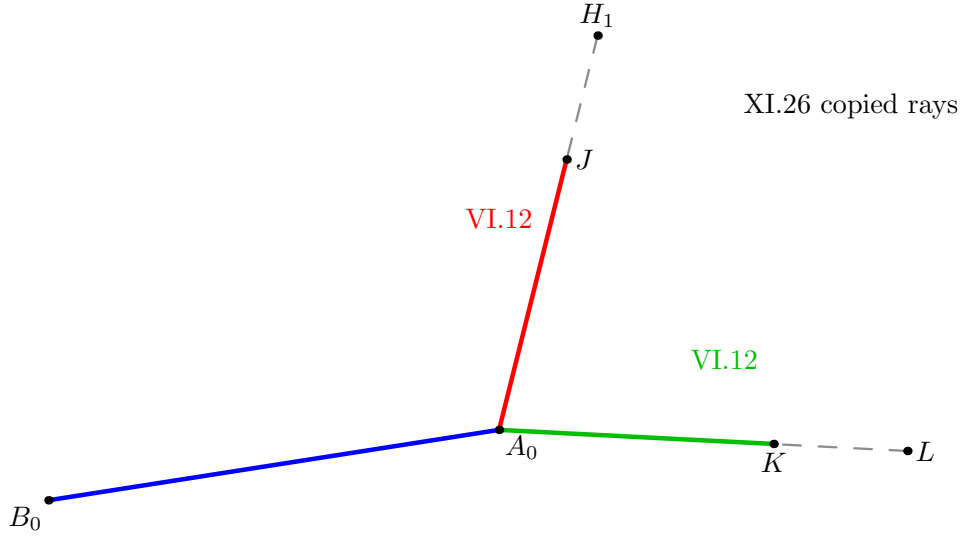
By V.22,

$$BA : BF = A_0B_0 : A_0J.$$

Hence the three incident target edges

$$A_0B_0, \quad A_0K, \quad A_0J$$

are synchronized by one common synthetic scale.



$$BA : BC = A_0B_0 : A_0K$$

V.22 synchronizes the outer ratio

$$BC : BF = A_0K : A_0J$$

$$BA : BF = A_0B_0 : A_0J$$

Figure 2: The copied trihedral corner at A_0 after the two fourth-proportional constructions. The points K and J lie on the rays copied by XI.26. VI.12 supplies the first two proportions and V.22 supplies the third.

Finally the three target edge directions are completed to a full parallelepipedal shell by translated parallel directions and opposite parallel planes. The three faces adjacent to A_0 are similar to the three corresponding source faces by proportional adjacent sides and equal included angles. The three opposite faces are then recovered from the parallel-shell structure.

4 Euclid's Classical Proof

Euclid's proof can be organized into six principal steps.

4.1 Copy the source solid angle

Using XI.26, Euclid constructs at the prescribed point on the prescribed straight line a solid angle equal to one corner of the given parallelepipedal solid.

In the classical notation this produces three equal plane face angles. In the formal proof it produces the trihedral configuration

```
HilbertTrihedralConfiguration Geo A0 B0 L H1
```

together with three angle congruences.

The first target edge is already fixed by the given straight line. Only the lengths on the other two copied rays remain to be determined.

4.2 Choose the second target edge by VI.12

Euclid constructs the fourth proportional corresponding to

$$BA : BC = A_0B_0 : A_0K.$$

The formal proof does not merely construct an abstract fourth segment. It constructs K on the already prescribed copied ray A_0L . This preserves both the required ratio and the copied angular configuration.

4.3 Choose the third target edge by VI.12

A second fourth-proportional construction gives J on the ray A_0H_1 with

$$BC : BF = A_0K : A_0J.$$

At this stage two of the three face ratios are available directly.

4.4 Synchronize the scale by V.22

Euclid now applies the first combination rule:

$$BA : BC = A_0B_0 : A_0K,$$

$$BC : BF = A_0K : A_0J,$$

hence

$$BA : BF = A_0B_0 : A_0J.$$

This is the decisive compatibility statement. Without it, the two successive fourth-proportional constructions would not yet imply that all three faces around the target corner use one consistent scale.

4.5 Complete the target parallelepiped

Through the three scaled edge directions Euclid completes the parallelepipedal solid.

The formal implementation isolates this as a reusable shell construction. Starting from a proper trihedral corner, translated edge directions produce the three opposite planes and then the remaining four vertices.

4.6 Identify the six corresponding faces

For each of the three faces incident with the chosen corner, the sides about the included angle are proportional and the included angles are equal. Therefore the corresponding parallelograms are similar.

The opposite faces of a parallelepiped are equal and similar to their adjacent counterparts. Consequently all six faces correspond, and by XI.Def.9 the two parallelepipedal solids are similar and similarly situated.

The formal proof makes the last step explicit through a six-face packaging predicate rather than leaving the passage from three adjacent faces to the whole solid implicit.

5 Why XI.27 Is Structurally Difficult

The classical proof is short because several compatibility statements are compressed into familiar geometric language. The formal reconstruction exposes four separate difficulties.

5.1 A copied solid angle is not yet a similar solid

XI.26 determines the three target directions but says nothing about the relative target edge lengths. A solid angle determines shape at the vertex, not scale along its rays.

The formal proof must therefore keep the XI.26 angular data intact while moving to new points K and J on the same rays.

5.2 The two fourth-proportional constructions must share one scale

The first VI.12 construction determines A_0K from BA, BC, A_0B_0 . The second determines A_0J from BC, BF, A_0K . Their compatibility is not definitional. V.22 is needed to derive the third relation

$$BA : BF = A_0B_0 : A_0J.$$

This is why XI.27 forced the production completion of the raw synthetic V.22 machinery.

5.3 The completed shell is genuinely spatial

The three scaled rays at A_0 determine only one trihedral corner. A full parallelepiped requires:

- translated edge directions;
- three opposite planes;
- four further vertices;
- all six face incidences and parallel relationships.

The shell construction is therefore a substantial spatial theorem, not a mere definition.

5.4 XI.Def.9 is a six-face statement

Euclid reasons from the three adjacent faces and the standard equality of opposite faces. In Lean the corresponding transport must be made explicit. The three opposite face similarities require both opposite-edge congruence and opposite-face angle congruence.

The latter is recovered using XI.10, while the former is proved face by face inside local plane geometry.

6 Formal Reconstruction: Source Corner and Scaling

6.1 Extracting the source trihedral corner

The source parallelepiped configuration supplies the trihedral corner at B :

```
hilbertParallelepipedConfiguration_trihedral_at_B
```

The resulting configuration is

```
HilbertTrihedralConfiguration Geo B A C F
```

and its three incident edge lengths are represented by

$$BA, \quad BC, \quad BF.$$

The nondegeneracy of these three edges is derived from the three proper face angles of the trihedral configuration.

6.2 XI.26 on the prescribed edge

The proof invokes

```
euclid_proposition_11_26
```

with source corner

$$B; A, C, F$$

and prescribed target edge

$$A_0B_0.$$

It returns points L, H_1 such that

```
HilbertTrihedralConfiguration Geo A0 B0 L H1
```

and the three copied face angles satisfy

$$\angle B_0A_0L \cong \angle ABC,$$

$$\angle LA_0H_1 \cong \angle CBF,$$

$$\angle H_1A_0B_0 \cong \angle FBA.$$

This is the angular skeleton of the target solid.

6.3 The first fourth proportional

The theorem

```
hilbertSpaceSegmentProportionRaw_fourth_on_ray_XI27
```

is the spatial wrapper used by XI.27 around the planar fourth-proportional construction.

A plane through B_0, A_0, L is chosen. Inside that plane the theorem constructs K on the same ray from A_0 as L and proves

$$BA : BC = A_0B_0 : A_0K.$$

The output includes both

```
HilbertSameRay Geo A0 L K
```

and the ambient raw proportion statement

```
HilbertSpaceSegmentProportionRaw
  Geo
  B A B C
  A0 B0 A0 K
```

6.4 The second fourth proportional

A second plane through L, A_0, H_1 is used for the next construction. The same wrapper produces J on the same ray as H_1 with

$$BC : BF = A_0K : A_0J.$$

Thus

```
HilbertSpaceSegmentProportionRaw
  Geo
  B C B F
  A0 K A0 J
```

is obtained while preserving the copied direction.

6.5 V.22 and the third proportion

The local interface

```
HilbertSpaceV22Raw_XI27
```

records the exact spatial raw combination rule required by XI.27:

$$a : b = a' : b', \quad b : c = b' : c' \implies a : c = a' : c'.$$

The theorem used here is not the quotient-level Eudoxus theorem

```
euclid_proposition_5_22
```

but the spatial raw theorem

```
hilbertSpaceSegmentProportionRaw_V22_of_crossing
```

from

```
CGJteamLab/Proposition5_22.lean
```

The XI.27 production file discharges the required plane-wise crossing hypothesis internally. It constructs

```
have hCross :  
  HilbertSpaceCrossingRaysTransfer  
    (Geo := Geo) := by  
intro pi  
exact  
  hilbert_crossing_rays_transfer  
    (PlaneGeo Geo pi)
```

Thus every plane slice receives the already closed planar theorem from Forder Book IV. The resulting theorem

```
hilbertSpaceV22Raw_XI27_of_crossing
```

packages the spatial raw V.22 rule for the XI.27 construction.

Applied to the two fourth-proportional relations, it yields

$$BA : BF = A_0B_0 : A_0J.$$

The three required edge proportions are therefore

$$\begin{aligned} BA : BC &= A_0B_0 : A_0K, \\ BC : BF &= A_0K : A_0J, \\ BA : BF &= A_0B_0 : A_0J. \end{aligned}$$

The role of V.22 is therefore precise: it does not choose a new point and it does not introduce a numerical scale factor. It proves that the two successive fourth-proportional constructions belong to one coherent three-edge scale.

7 Formal Reconstruction: Transport and Shell

7.1 Transporting the trihedral corner along the copied rays

The points K and J lie on the copied XI.26 rays, but the original trihedral configuration was stated using L, H_1 . The theorem

```
hilbert_XI27_trihedral_transport_sameRays
```

moves the proper trihedral configuration from

$$(A_0; B_0, L, H_1)$$

to

$$(A_0; B_0, K, J).$$

The result is

```
HilbertTrihedralConfiguration Geo A0 B0 K J
```

The corresponding angle equalities are transported by

```
hilbert_XI27_angleCongruent_sameRays_space
```

Hence the scaled target corner retains all three copied XI.26 angles.

7.2 The scaled-shell theorem

The central integration theorem is

```
hilbert_XI27_scaled_shell_exists
```

Conceptually it returns:

- the scaled target corner $A_0; B_0, K, J$;
- three copied angle congruences;
- three raw segment proportions;
- six target face planes;
- four additional target vertices;

- a complete target `HilbertParallelepipedConfiguration`.

The version

```
hilbert_XI27_scaled_shell_exists_of_crossing
```

discharges the raw V.22 obligation internally from the production crossing-rays theorem.

7.3 The parallelepiped shell

The folded production support begins from the theorem

```
hilbert_parallelepiped_shell_exists_from_trihedral
```

Its input is a proper trihedral corner. It constructs the opposite-plane shell by first producing translated edge directions and then the three opposite planes.

The main reusable stages are represented by

```
hilbert_trihedral_two_distinct_translates_at_A
hilbert_trihedral_opposite_plane_through_A
hilbert_trihedral_three_opposite_planes
hilbert_XI27_parallel_translates_distinct
hilbert_XI27_parallel_translates_meet
```

The final shell contains the eight target vertices

$$B_0, A_0, K, TD, TE, J, TG, TH$$

and six planes

$$t\pi_0, t\pi_1, \quad t\rho_0, t\rho_1, \quad t\sigma_0, t\sigma_1.$$

8 Formal Reconstruction: Face Similarity

8.1 The local face-corner relation

The proposition file introduces

```
HilbertSpaceFaceCornerSimilar
```

For triples

$$A, B, C \quad \text{and} \quad A', B', C',$$

the relation records two pieces of data:

1. the two incident sides about B and B' occur in the same raw synthetic proportion;

XI.Def.9: six corresponding face similarities

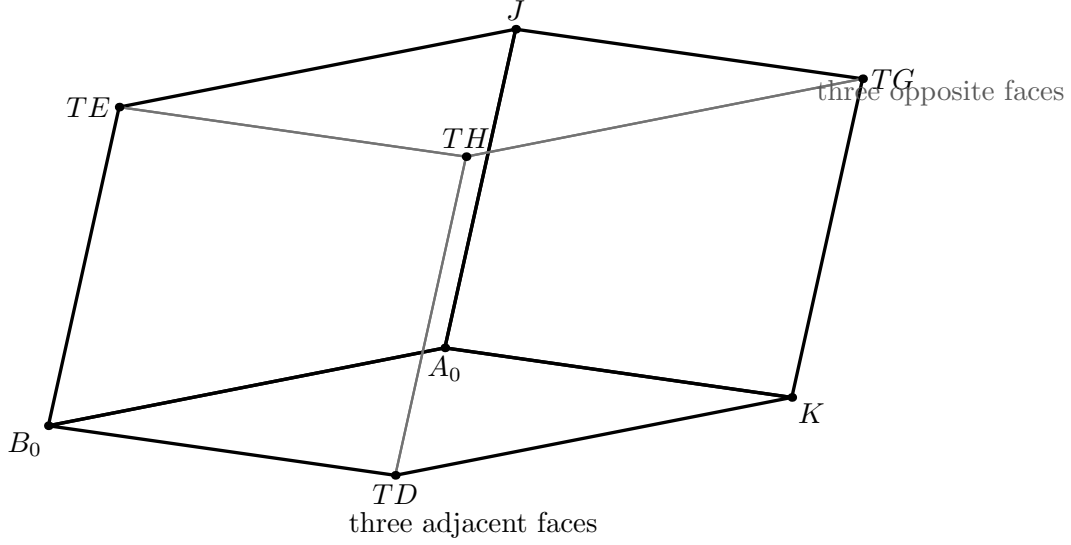


Figure 3: The completed target parallelepipedal shell. The three faces incident with A_0 are obtained directly from the scaled corner. Their three opposite faces are recovered by parallel-plane and opposite-edge transport before the final XI.Def.9 packaging.

2. the included angles at B and B' are congruent.

This is exactly the local SAS-style information needed for similarity of a parallelogram face.

8.2 The three adjacent faces

The theorem

hilbert_XI27_three_adjacent_face_corners_similar

packages the three angle congruences and three edge proportions into similarity data for the three faces incident with the chosen corner.

The source face corners are

$$(A, B, C), \quad (C, B, F), \quad (A, B, F),$$

and the corresponding target corners are

$$(B_0, A_0, K), \quad (K, A_0, J), \quad (B_0, A_0, J).$$

This is the direct formal counterpart of Euclid's statement that the three parallelograms adjacent to the chosen source corner are similar to the three parallelograms adjacent to the target corner.

8.3 Opposite edges without ambient global order

For a single face, opposite-edge congruence is established by

```
hilbert_XI27_face_opposite_sides_congruent
```

The important architectural feature is that the parallelogram argument is performed inside

```
PlaneGeo Geo pi
```

rather than by installing a global ambient `HilbertOrder Geo`.

The theorem

```
hilbert_XI27_opposite_face_edge_congruences
```

then collects the required opposite-edge equalities for all three face pairs.

8.4 Opposite face angles

The local directed-side relation for a parallelogram is provided by

```
hilbert_XI27_plane_parallelogram_directed_opposite_sides
```

The theorem

```
hilbert_XI27_opposite_pi_face_angle
```

uses the spatial parallel-line angle transport available from XI.10 to recover the included angle on one opposite face.

The combined result

```
hilbert_XI27_all_opposite_face_angles
```

provides the corresponding angle congruences for all three opposite face pairs.

8.5 The three opposite faces

The general transport theorem

```
hilbertSpaceFaceCornerSimilar_transport
```

moves a face-corner similarity through congruent replacements of the four incident sides and through congruent replacement of the included angle.

Using the opposite-edge and opposite-angle data, the theorem

```
hilbert_XI27_three_opposite_faces_similar
```

proves the similarity of the remaining three corresponding faces.

Thus all six face similarities required by XI.Def.9 are explicit.

9 XI.Def.9 Packaging

The project representation of a parallelepiped's six relevant face corners is

```
HilbertSpaceFaceCorner
```

with the face-similarity relation

```
HilbertSpaceFaceCornerSimilarFace
```

The six corners are assembled by

```
hilbertXI27FaceCorners
```

The final packaging theorem is

```
hilbert_XI27_XI9_packaging
```

It takes:

- the source parallelepiped configuration;
- the target parallelepiped configuration;
- similarity of the three adjacent face corners.

It derives the three opposite face similarities and returns

```
HilbertXI9SimilarParallelepiped  
  (HilbertSpaceFaceCornerSimilarFace  
   (Geo := Geo))  
  ...
```

The public theorem

```
euclid_proposition_11_27
```

therefore has a short final proof: construct the scaled shell, build the three adjacent similarities, invoke the XI.Def.9 packaging theorem, and return the target solid.

10 Proof Architecture and Formal Obligations

The complete formal reconstruction can be read as the following continuous construction.

1. Extract the source trihedral corner $B; A, C, F$ from the source parallelepiped.
2. Apply XI.26 to copy that trihedral angle at A_0 on the prescribed ray A_0B_0 , obtaining rays A_0L and A_0H_1 .
3. In the first copied face plane, use VI.12 to choose K on A_0L with

$$BA : BC = A_0B_0 : A_0K.$$

4. In the second copied face plane, use VI.12 to choose J on A_0H_1 with

$$BC : BF = A_0K : A_0J.$$

5. Use V.22 to obtain

$$BA : BF = A_0B_0 : A_0J.$$

6. Transport the XI.26 trihedral configuration and the three copied angles from L, H_1 to K, J along the same rays.
7. Complete the scaled trihedral corner to a full target parallelepipedal shell.
8. Convert the three angle/proportion pairs into similarity of the three adjacent face corners.
9. Recover opposite-edge congruences in local plane geometry.
10. Recover opposite-face angles by spatial parallel-angle transport.
11. Transport the three adjacent similarities to the three opposite faces.
12. Package the six corresponding face similarities as `HilbertXI9SimilarParallelepiped`.

Several obligations which are nearly invisible in Euclid’s text become explicit in Lean:

- the two fourth-proportional points must lie on the already copied rays, not merely on arbitrary congruent rays;
- the two fourth-proportional constructions must be synchronized by a genuine V.22 theorem;
- the scaled target rays must still form a proper trihedral corner;
- the translated edge directions in the shell construction must be distinct and must meet in the required planes;
- all eight target vertices must satisfy one common `HilbertParallelepipedConfiguration`;
- opposite-side equalities of a parallelogram must be recovered without imposing unnecessary global ambient plane order;
- three adjacent face similarities are not by themselves the project representation of XI.Def.9; the three opposite faces must be supplied explicitly.

The formal length of XI.27 is therefore the cost of exposing the scale, incidence, orientation, and six-face compatibility assumptions hidden in the classical phrase “complete the parallelepiped.”

11 Dependencies and Reuse

The production XI.27 file imports a substantial but structured dependency set.

The principal Book XI dependencies are

```
CGJteamLab.Proposition11_10
CGJteamLab.Proposition11_12
CGJteamLab.Proposition11_15
CGJteamLab.Proposition11_16
CGJteamLab.Proposition11_24
CGJteamLab.Proposition11_25
CGJteamLab.Proposition11_26
```

Their roles are different:

- XI.26 supplies the copied trihedral angle;
- XI.24 supplies the source parallelepiped corner infrastructure;
- XI.10 is used in opposite-face angle transport;
- XI.12, XI.15, XI.16, and the spatial parallel infrastructure support the parallelepiped shell construction;
- XI.25 remains part of the established proportion/similarity infrastructure used by the consolidated development.

The proportion layer is supplied by

```
CGJteamLab.Hilbert3DProportion
CGJteamLab.HilbertInterfaceVI
CGJteamLab.Proposition5_22
```

The circle and crossing-rays development needed for the raw synthetic V.22 route is now productionized in the split Book IV modules. The public endpoint actually consumed by V.22 is

```
CGJteamLab.Proposition4_20
```

together with its proposition dependency closure and the reusable

```
CGJteamLab.HilbertInterfaceIV
```

XI.27 itself has no direct import from Book IV. Its direct dependency is `Proposition5_22`. The crossing endpoint consumed there is

```
hilbert_crossing_rays_transfer
```

from `Proposition4_20`.

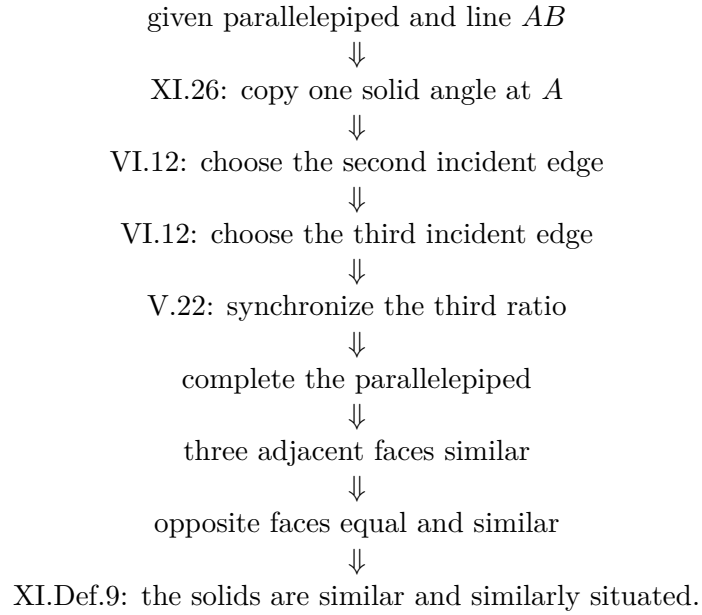
This separation is important. The final XI.27 production graph no longer imports workshop files named

```
Crossing_circle_stage*
V22_*
XI27_*
```

and there is no remaining aggregate `PropositionHilbertIV.lean`. The corresponding mathematics is owned by the proposition files IV.12–IV.20 and their interface module.

12 Classical and Formal Dependency DAGs

The classical construction is compact:



The formal production chain is:

HilbertParallelepipedConfiguration
 $\Downarrow$
 trihedral corner at B
 $\Downarrow$
 euclid_proposition_11_26
 $\Downarrow$
 L, H_1 with three copied angles
 $\Downarrow$
 hilbertSpaceSegmentProportionRaw_fourth_on_ray_XI27
 $\Downarrow$
 K, J with two raw proportions
 $\Downarrow$
 Proposition5_22
 $\Downarrow$
 third raw proportion
 $\Downarrow$
 hilbert_XI27_trihedral_transport_sameRays
 $\Downarrow$
 hilbert_parallelepiped_shell_exists_from_trihedral
 $\Downarrow$
 three adjacent face similarities
 $\Downarrow$
 opposite edges + XI.10 opposite angles
 $\Downarrow$
 three opposite face similarities
 $\Downarrow$
 hilbert_XI27_XI9_packaging
 $\Downarrow$
 euclid_proposition_11_27.

The most important architectural addition relative to Euclid's terse proof is the explicit intermediate chain

Forder IV.12–IV.20
 $\Downarrow$
 hilbert_crossing_rays_transfer
 $\Downarrow$
 raw alternando
 $\Downarrow$
 spatial raw V.22
 $\Downarrow$
 XI.27 scale synchronization

The planar crossing theorem is closed in `Proposition4_20.lean`; XI.27 then lifts it plane by plane to the ambient spatial setting through `PlaneGeo`.

13 Comparison with Euclid

At the conceptual level the formal proof follows Euclid very closely.

1. Copy the solid angle of the given parallelepiped to the prescribed line.
2. Choose the remaining two incident target edges by proportional construction.
3. Use V.22 to show that all three target edges use one consistent scale.
4. Complete the target parallelepiped.
5. Prove the three adjacent faces similar.
6. Transfer similarity to the three opposite faces.
7. Invoke the definition of similar parallelepipedal solids.

The differences are mainly representational and architectural.

First, Euclid speaks directly about similar parallelograms. The formal proof uses the local predicate

```
HilbertSpaceFaceCornerSimilar
```

which records the proportional adjacent sides and the congruent included angle needed to represent the same synthetic information.

Second, Euclid’s phrase “complete the parallelepiped” suppresses a substantial spatial construction. The formal proof explicitly creates translated edge directions, three opposite planes, and four remaining vertices.

Third, Euclid uses the familiar fact that opposite faces of a parallelepiped are equal and similar. The Lean theorem decomposes this into opposite-edge congruence, opposite-face angle congruence, and a transport theorem for face similarity.

Fourth, the formal proof deliberately avoids a global ambient order structure when only plane-local order is needed. This makes the dependency profile of XI.27 more faithful to the actual geometry used in each step.

Finally, the formal development exposes an important hidden dependency of the classical proof: the two successive fourth-proportional constructions become one coherent three-edge scale only after a genuine V.22 theorem is available.

14 Diagrammatic Audit

Three figures are mathematically justified for the final documentation.

1. **Source parallelepiped and selected corner.** Show the source solid and emphasize the corner at B with incident edges BA, BC, BF . The opposite faces should be visible but secondary.
2. **Copied and scaled trihedral corner.** Show the prescribed edge A_0B_0 , the original copied XI.26 rays A_0L, A_0H_1 , and the scaled points K, J on those rays. The figure should communicate the two VI.12 constructions and the fact that V.22 supplies the third ratio.
3. **Completed target shell and corresponding faces.** Show the eight target vertices and distinguish the three faces adjacent to A_0 from the three opposite faces. The figure should support the XI.Def.9 discussion rather than carry metric labels.

The proposed production filenames are

```
figures/xi27_fig01_source_parallelepiped_corner.pdf
figures/xi27_fig02_scaled_triheral_corner.pdf
figures/xi27_fig03_target_shell_similarity.pdf
```

No separate figure is required for the crossing-circle proof of V.22 inside the XI.27 document. That development is mathematically reusable and should be documented in its own Hilbert/Forder IV and V.22 documentation rather than turning the XI.27 report into a second proof of the proportion theory.

15 Audit Status

This revision is aligned with the current production repository after the Book IV split. The authoritative XI.27 source is

```
CGJteamLab/Proposition11_27.lean
```

at repository blob

```
c9400a5195a00e52153a8fe2dcd8971a55ad6b89
```

The production sources most relevant to the present documentation are

```
CGJteamLab/HilbertInterfaceIV.lean
CGJteamLab/Proposition4_12.lean
...
CGJteamLab/Proposition4_20.lean
CGJteamLab/Proposition5_22.lean
CGJteamLab/HilbertInterfaceVI.lean
CGJteamLab/HilbertInterfaceXI.lean
CGJteamLab/Proposition11_24.lean
CGJteamLab/Proposition11_26.lean
CGJteamLab/Proposition11_27.lean
```

The direct import list of `Proposition11_27.lean` contains `Proposition5_22` but no Book IV proposition. The circle dependency enters transitively:

$$\text{Proposition11_27} \leftarrow \text{Proposition5_22} \leftarrow \text{Proposition4_20}.$$

The former aggregate

```
CGJteamLab/PropositionHilbertIV.lean
```

is not part of the current production repository. Its mathematical content has been distributed between `HilbertInterfaceIV.lean` and the numbered proposition modules IV.12–IV.20, ending at `Proposition4_20.lean`.

The current XI.27 source also contains the folded spatial integration theorem

```
hilbertSpaceV22Raw_XI27_of_crossing
```

which constructs the plane-wise `HilbertSpaceCrossingRaysTransfer` directly from the closed planar theorem.

This documentation does not claim a new kernel

```
#print axioms Geometry.euclid_proposition_11_27
```

audit unless that command is run and recorded separately. Likewise, build status should be read from the current Lean repository rather than inferred from this documentation snapshot.

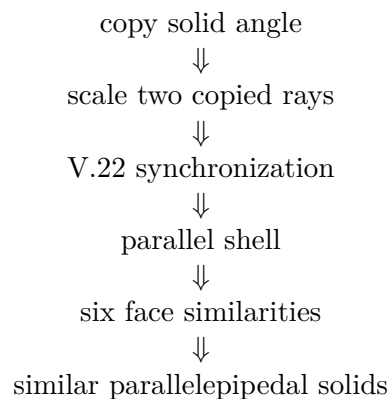
The current production route is the direct Hilbert 3D route. No separate XI.27 Wyler proposition is asserted in this document.

16 Conclusion

Proposition XI.27 is the point at which Euclid’s spatial congruence theory and his theory of proportion meet in a constructive theorem about similar solids.

XI.26 supplies the target directions. VI.12 determines two target edge lengths on those directions. V.22 proves that the three incident target edges share one coherent scale. A spatial parallel construction completes the target parallelepiped, and XI.Def.9 packages the six corresponding face similarities.

The production formalization therefore has the structural form



The main formal lesson is that “similar and similarly situated” is not a single primitive numerical relation. In the synthetic development it is assembled from incidence, ray orientation, copied angles, fourth proportionals, proportion composition, translated parallel directions, and face-by-face similarity transport.

This preserves the classical mechanism while making explicit the logical content hidden by Euclid’s compact construction.

XI.27 also marks an architectural consolidation point for the project. The circle geometry required by raw alternando is now owned by the split Forder IV.12–IV.20 production modules, with

`hilbert_crossing_rays_transfer` exported from IV.20. `Proposition5_22.lean` converts that planar result into raw planar and spatial V.22, and XI.27 consumes the spatial form plane by plane.

The final theorem therefore records not only the completion of one Euclidean construction but also a clean production boundary between Forder Book IV circle geometry, Book V proportion theory, Book VI proportional construction, and Book XI solid geometry.

References

- [1] Euclid, *The Thirteen Books of Euclid's Elements*, edited and translated by Sir Thomas L. Heath, 3 vols., Dover Publications, New York, 1956. (Reprint of the second edition, Cambridge University Press, 1926.)